Gideon A. **Afriyie**

Mathematics Undergraduate | Gettysburg College

https://gideonafriyie.github.io @ gideonafriyie23@gmail.com linkedin.com/in/gideon-afriyie Google Scholar

Education

May 2027 | **Gettysburg College - B.A. in Mathematics** — **Gettysburg, PA, USA**
Aug 2023 | AMS F-GAP Scholar, Taylor Gaw '13 Endowed Fellowship, Seibert Project Award

Experience

Present | **Gettysburg College | Advancing Science Lab** [🌐] — **Gettysburg, PA**
Aug 2026 | *Research Assistant*

- Developing tactile 3D topographic models of South Mountain for K–12 geoscience education.
- Integrated elevation, transportation, hydrography, and trail datasets using QGIS and Python, applying spatial filtering, clipping, and coordinate reprojection.
- Designed and tested 3D-printed terrain prototypes featuring raised roads, recessed waterways, and the Appalachian Trail.

Jul 2025 | **Under Canvas Inc.** — **Remote**
May 2025 | *Data Analytics Fellow*

- Analyzed a $1.6M employee compensation dataset using Python and Excel to identify anomalies across employee groups.
- Developed quantitative recommendations for incentive restructuring and presented findings.

Publications

[J.1] **Extreme Value Theory Analysis of Prime Gap Distributions: Statistical Analysis of Cramér's Conjecture and Light-Tailed Behavior** [DOI: 10.1016/j.spl.2026.110898]
Gideon A. Afriyie
Journal: Statistics & Probability Letters (Elsevier) [**2026**]

[P.1] **Predicting Urban Air Quality Trends Using Machine Learning and Public Satellite/Sensor Data** [Link]
R. Haque, B. Kebede, G. A. Afriyie, H. Yusuf
Summer Proceedings 2026, Chapter 16 [**2026**]

Programs & Training

Stanford University | Probability for AI [🌐] — Oct 2026 – Nov 2026
Program Participant

- Exploring probabilistic foundations of AI through applied projects in statistical inference, ML, and model evaluation.

MIT/Harvard | Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) [🌐] — Aug 2026
AI Summer School

- Studied symbolic regression, diffusion models, and simulation-based inference; participated in an AI hackathon.

Google | TPU Research Cloud [🌐] — May 2026
Research Program

- Selected for the TPU Research Cloud program, providing access to Google Cloud TPUs for machine learning research.

Projects

Quantitative Risk & Portfolio Optimization [GitHub]

- Analyzed 6+ years of price data across 8 assets, estimating returns, volatility, and correlations (VOO–VIXY: -0.754).
- Optimized portfolios using mean-variance analysis and efficient frontiers, applying KKT to restrict short selling.

Skills

Python, Excel, SQL, LaTeX